

CSE402: Optimisation

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1 Lecture 01

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Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

§1.1 Review on linear algebra

Notation. Let $a_1, a_2, \dots, a_n$ be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \dots + a_n \quad (1.1)$$

by

$$S = \sum_{k=1}^n a_k. \quad (1.2)$$

We have the following useful identities:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad (1.3)$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}. \quad (1.4)$$

§1.1.1 Vectors

Definition 1.1.1 ($\mathbb{R}$ -vector space). Let V be a non-empty set. The set V equipped with two binary operations

$$+ : V \times V \rightarrow V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \rightarrow V$$

called *scalar multiplication* is called a **vector space over $\mathbb{R}$** or **$\mathbb{R}$ -vector space** if for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. **Closure under addition:**

$$u + v \in V \quad (1.5)$$

2. **Commutativity of addition:**

$$u + v = v + u \quad (1.6)$$

3. **Associativity of addition:**

$$(u + v) + w = u + (v + w) \quad (1.7)$$

4. **Existence of zero vector:** There exists a vector $0 \in V$ such that

$$u + 0 = u \quad (1.8)$$

5. **Existence of additive inverse:** For every $u \in V$, there exists $-u \in V$ such that

$$u + (-u) = 0 \quad (1.9)$$

6. **Closure under scalar multiplication:**

$$au \in V \quad (1.10)$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \quad (1.11)$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \quad (1.12)$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \quad (1.13)$$

10. **Identity element of scalar multiplication:**

$$1u = u \quad (1.14)$$

Hence, $(V, +, \cdot)$ is called a vector space over the field $\mathbb{R}$. Elements of V are called **vectors** and elements of $\mathbb{R}$ are called **scalars**.

Convention. From now on, vector space means $\mathbb{R}$ -vector space, unless stated otherwise.

In this course, we will work with only the following special example of $\mathbb{R}$ -vector space.

Example 1.1.2

Let

$$V := \mathbb{R}^n = \{[x_1, x_2, \dots, x_n]^T \mid x_1, x_2, \dots, x_n \in \mathbb{R}\}. \quad (1.15)$$

For $[x_1, x_2, \dots, x_n]^T, [y_1, y_2, \dots, y_n]^T \in V$ and $\alpha \in \mathbb{R}$ define addition (+) and scalar multiplication ($\cdot$), respectively as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad (1.16)$$

$$\alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \quad (1.17)$$

Then it is an easy check that $V = \mathbb{R}^n$ forms an $\mathbb{R}$ -vector space.

§1.1.2 Linear transformations

In this course, we are mostly concerned with following special functions between vector spaces:

Not sure about that

Definition 1.1.3 (Linear transformation). Let V, W be two vector space. A **linear transformation** from V to W is a function $T : V \rightarrow W$ such that for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathbb{R}$ the following holds true:

- **Additivity.**

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}). \quad (1.18)$$

- **Homogeneity.**

$$T(\alpha \cdot \mathbf{x}) = \alpha \cdot T(\mathbf{x}). \quad (1.19)$$

Need to add visualisation of additivity and homogeneity property of a linear transformation.

Example 1.1.4

Let $V := \mathbb{R}^n, W := \mathbb{R}^m$. An $m \times n$ matrix M then defines a linear transformation $M : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ^a as follows:

For $\mathbf{x} \in \mathbb{R}^n$,

$$M(\mathbf{x}) := M\mathbf{x}. \quad (1.20)$$

Here, M is an $m \times n$ matrix, $\mathbf{x}$ can is an $n \times 1$ matrix. So $M\mathbf{x}$ is an $m \times 1$ matrix. Therefore, M is indeed a function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Furthermore, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, the following holds by the properties of matrix multiplication:

$$M(\mathbf{x} + \mathbf{y}) = M\mathbf{x} + M\mathbf{y}, \quad (1.21)$$

$$M(\alpha\mathbf{x}) = \alpha M\mathbf{x} \quad (1.22)$$

Hence, M is indeed a linear transformation.

^aHere, we are denoting the linear transformation determined by the matrix M by the same letter M using abuse of notation.

Surprising fact is that an arbitrary linear transformation also comes from a matrix. Before that, we need the idea of basis.

§1.1.3 Basis and dimension

Definition 1.1.5 (Linear combination). Let V be a vector space, and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in V$. A **linear combination** of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is an expression of the form

$$c_1\mathbf{x}_1, c_2\mathbf{x}_2, \dots, c_n\mathbf{x}_n, \quad (1.23)$$

where $c_1, c_2, \dots, c_n \in \mathbb{R}$.

§1.1.4 Eigenvectors and eigenvalues

Suppose we have a linear transformation $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We are interested in finding a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda\mathbf{x}, \quad (1.24)$$

where $\lambda \in \mathbb{R}$.

Definition 1.1.6 (Eigenvector and eigenvalue). $\mathbf{x}, \lambda$ (if exists) are called an **eigenvector** and **eigenvalue** for L . The $(\mathbf{x}, \lambda)$ is called an **eigenpair** for L .

Theorem 1.1.7

Let L be a linear transformation. Then $(\mathbf{x}, \lambda)$ is an eigen pair of L if and only if $\det(L - \lambda I) = 0$.

Proof. Suppose λ is an eigenvalue of L . Then there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda\mathbf{x}. \quad (1.25)$$

Since $I\mathbf{x} = \mathbf{x}$, we have

$$L\mathbf{x} - \lambda I\mathbf{x} = 0, \quad (1.26)$$

or equivalently,

$$(L - \lambda I)\mathbf{x} = 0. \quad (1.27)$$

Since $\mathbf{x} \neq 0$, this homogeneous system has a non-trivial solution. Therefore, $L - \lambda I$ is singular. Hence,

$$\det(L - \lambda I) = 0. \quad (1.28)$$

Conversely, suppose $\det(L - \lambda I) = 0$. Then $L - \lambda I$ is singular, so there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ satisfying

$$(L - \lambda I)\mathbf{x} = 0. \quad (1.29)$$

Therefore,

$$L\mathbf{x} = \lambda\mathbf{x}. \quad (1.30)$$

Thus, λ is an eigenvalue of L , with corresponding eigenvector $\mathbf{x}$. $\square$

Definition 1.1.8 (Eigenvalues of a matrix). Let M be an $n \times n$ matrix. The set of all eigenvalues of M is denoted by

$$\text{eigs}(M) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}. \quad (1.31)$$

Here, the eigenvalues are listed with multiplicity.

Definition 1.1.9 (Definiteness of a matrix). Let M be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Then M is called

1. **Positive definite (PD)** if all eigenvalues are strictly positive, i.e.,

$$\lambda_i > 0, \quad \forall i = 1, 2, \dots, n. \quad (1.32)$$

2. **Negative definite (ND)** if all eigenvalues are strictly negative, i.e.,

$$\lambda_i < 0, \quad \forall i = 1, 2, \dots, n. \quad (1.33)$$

3. **Positive semidefinite (PSD)** if all eigenvalues are non-negative, i.e.,

$$\lambda_i \geq 0, \quad \forall i = 1, 2, \dots, n. \quad (1.34)$$

4. **Negative semidefinite (NSD)** if all eigenvalues are non-positive, i.e.,

$$\lambda_i \leq 0, \quad \forall i = 1, 2, \dots, n. \quad (1.35)$$

5. **Indefinite (ID)** if M has both positive and negative eigenvalues.

Example 1.1.10

Let

$$L := \begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}. \quad (1.36)$$

We will find the eigenvalues and eigenvectors for L .

The eigenvalues are obtained by solving $\det(L - \lambda I) = 0$. Here,

$$\begin{aligned} \det(L - \lambda I) &= \begin{vmatrix} -2 - \lambda & 2 \\ 3 & -1 - \lambda \end{vmatrix} \\ &= (-2 - \lambda)(-1 - \lambda) - 6 \\ &= \lambda^2 + 3\lambda - 4 \\ &= (\lambda - 1)(\lambda + 4). \end{aligned} \quad (1.37)$$

Hence, the eigenvalues for L are $\lambda = 1$ and $\lambda = -4$.

For each eigenvalue, we find a non-zero vector $\mathbf{x} = [x_1, x_2]^T$ such that $(L - \lambda I)\mathbf{x} = \mathbf{0}$.

For $\lambda = 1$, we have

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.38)$$

Both equations give $3x_1 = 2x_2$, so $x_2 = \frac{3}{2}x_1$. Therefore, the eigenvectors corresponding to $\lambda = 1$ are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \frac{3}{2}x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ \frac{3}{2} \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\}. \quad (1.39)$$

Choosing $x_1 = 2$, we obtain the eigenvector $\mathbf{x} = [2, 3]^T$.

For $\lambda = -4$, we have

$$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.40)$$

Both equations give $x_1 + x_2 = 0$, so $x_2 = -x_1$. Therefore, the eigenvectors corresponding to $\lambda = -4$ are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\}. \quad (1.41)$$

Choosing $x_1 = 1$, we obtain the eigenvector $\mathbf{x} = [1, -1]^T$.

Hence, $([2, 3]^T, 1)$ and $([1, -1]^T, -4)$ are eigenpairs for L .

Remark 1.1.11. Any nonzero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.

Indeed, let $x \neq 0$ be an eigenvector of L corresponding to the eigenvalue λ . Then

$$Lx = \lambda x. \quad (1.42)$$

For any scalar $c \neq 0$, by linearity of L ,

$$L(cx) = cLx = c\lambda x = \lambda(cx). \quad (1.43)$$

Hence, cx is also an eigenvector of L with the same eigenvalue λ .